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Paper Code : 21325
F-425
B.C.A. (Fifth Semester)
Examination, 2025-26
(New Course)
Paper - BCA-505-N
ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Explain the need for parallelism in modern computer systems.

   (b) Discuss different models of parallel computation and their relevance in current architectures.

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2. (a) Compare cache write strategies: Write-through, Write-back, and Write-around.

   (b) Explain the different memory addressing schemes used in parallel computers.

3. (a) Illustrate the stages of an Instruction Pipeline with a neat diagram.

   (b) What is pipeline interlocking? Explain methods used to handle interlocks.

4. (a) Explain the architecture and working of a Vector Processing system with suitable examples.

   (b) Discuss Vector Chaining and its impact on system performance.

5. (a) Write short notes on the following (any two):

   1. Shuffle Exchange Network
   2. SIMT Architecture
   3. Barrel Shifter

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(b) Explain the algorithms used for matrix addition in SIMD processors.

6. (a) Differentiate between centralized and distributed shared memory models.

(b) Explain the structure and role of Inter-connection Networks in multiprocessors.

7. (a) Describe the UMA, NUMA, and NORMA memory models with their characteristics

(b) Explain tightly coupled multiprocessors and discuss their advantages.

8. Write in brief about the following (any four):

(a) Out-of-order execution

(b) VLIW architecture

(c) Snoopy cache protocol

(d) Flynn's taxonomy

(e) Multiprocessor scheduling

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Paper Code : 21325
F-425
B.C.A. (Fifth Semester)
Examination, 2023-24
(New Course)

Paper - BCA-505-N

ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are data hazards that are noticed in instruction level parallelism? Explain dynamic scheduling for overcoming these hazards.

(b) What is associative memory? Explain its advantages. 7+7

2. (a) An analysis requires multiplication of very large matrices. Which type of computer will you suggest for them? Give reason in support of you answer.

(b) Differentiate between data flow computer and control flow computer. 7+7

3. What is the major hurdle of pipelining? How do they reduce the performance? What are the techniques to be used to improve the performance? 14

4. What are the five primary approaches, in use, for multiple issue processors? Give the primary characteristics that distinguish them. 14

5. What is multiprocessor cache coherence? Illustrate the problem and show how different processors have different values for the same location. 14

6. (a) What is distributed memory multicomputer?

   (b) Explain SIMD computers? 7+7

7. Explain these terms with diagrams (if any) : 14

   (i) RISC scaler processors

   (ii) Loosely coupled multiprocessors

   (iii) UMA

   (iv) MUMA

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Paper Code : 21325
F-425

B.C.A. (Fifth Sem.) Examination, 2022-23

(New Course)

Paper-BCA-505-N

ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours] ***[Maximum Marks : 70***

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Explain parallelism in Uniprocessor systems.

   (b) Describe data driven computing with example.

2. (a) What are various addressing schemes? Explain with example.

   (b) Explain reservation Table Design for Instruction pipelined units.

3. (a) What is SIMD? Describe suffle exchange interconnection network.

   (b) What is job sequencing? Explain.

4. (a) What is multiprocessing? Differentiate loosely coupled multiprocessors with Tightly coupled multiprocessors.

   (b) What is Input-output subsystem? Describe with proper example?

5. (a) What is difference between MUMA and Time shared Interconnection Network.

   (b) State and explain parallel sorting algorithm for array processor.

6. (a) What are multiprocessor scheduling strategies? Explain.

(b) What do you understand SPARC architecture beneficial for parallelism.

7. (a) What is Window register concept? State your comments.

(b) Describe Architecture of TI-ASC system.

8. Explain any **two** of the following :

(a) Multiport Memories.

(b) Addressing schemes for main memory.

(c) RISC Scalar Processors

(d) Matrix Multiplication on Concurrent Processor

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Paper Code : 21325

F-425

B.C.A. (Fifth Semester) Examination, 2021

(New Course)

Paper No. BCA-505-N

ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What do you understand by parallel computer structures? Explain.

   (b) Explain the difference between control flow and data flow computers.

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2. (a) Explain the characteristics of cache memory in detail.

   (b) Explain the characteristics of input output subsystems in detail.

3. (a) Explain the principles of linear pipelining in detail.

   (b) Explain the characteristic of vector processing. https://www.mjpruonline.com

4. (a) What are the Parallel algorithms for array processors? Explain in detail.

   (b) How can we prevent collision in pipelining processing.

5. (a) What is the difference between loosely coupled and tighty coupled multiprocessors? Explain in detail.

   (b) Explain the classification of multiprocessor operating systems in detail.

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6. (a) Explain multiprocessor scheduling strategies in detail.

   (b) Explain SPARC architecture in detail.

7. (a) Explain superscalar processors in detail? What are the characteristic of it?

   (b) What are the difference between UMA and COMA? Explain in detail.

8. Explain any three terms:

   (i) Star-100 Architecture

   (ii) MUMA

   (iii) Parallel Sorting

   (iv) Crossbar Switch

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Total No. of Questions : 7] [Total No. of Printed Pages : 3

Paper Code : 21325

F–425

B.C.A. (Vth Semester) Examination, 2019–20

(New Course)

ADVANCED COMPUTER ARCHITECTURE

Paper-BCA-505-N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are the limitations of instruction-level parallelism ? Explain briefly.

   (b) Explain the four machine organizations according to Flynn's classification.

   (c) Briefly explain the alternative flow control strategies in any message passing system. 4,4,6

2. (a) What are different addressing techniques of cache memory ? Explain with example.

   (b) How cache memory and virtual memory are different from main memory of the processor ? Explain the significance of each memory type. 8,6

3. (a) Differentiate between linear pipeline and non-linear pipeline processor.

   (b) Distinguish between arithmetic and logical shifts. Show that in the former case signs are preserved.

   (c) Vectorizing compilers generally defect loops that can be executed on a pipelined vector computer. Are the vectorization algorithms used by vectorizing compilers suitable for MIMD machine parallelization ? Justify your answer. 4,4,6

4. (a) What are SIMD array processors ? Explain with suitable example and figure how communication takes place among the processing elements.

   (b) Explain hypercube processor organization. 9,5

5. (a) Differentiate between Loosely Coupled System (LCS) and Tightly Coupled System (TCS) and state which of the coupling systems is better for higher degree of interactions between tasks.

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(b) Explain the difference among the UMA, NUMA and COMA computer memory organisation models.

(c) Explain the crossbar switch interconnection network. 4,6,4

6. (a) Explain briefly how RISC architecture attempts to reduce execution time.

(b) What is VLIW (Very Large Instruction Word) and how is it different from RISC or CISC ?

(c) Suggest how VLIW architecture can achieve superscalar performance. 4,4,6

7. Write short notes on the following :

(a) Speedup, latency and Throughput

(b) Moore's law

(c) Iron law of processor performance

(d) Amdahl's law for vectorization 3,3,3,5

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Paper Code : 21325

F–425

B.C.A. (Fifth Semester) Examination, 2018

(New Course)

Paper-BCA-505-N

ADVANCED COMPUTER ARCHITECTURE

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) How can one achieve parallelism in Uniprocessor system ? Explain in brief.

   (b) What are various parallel computer system ? Explain.

2. (a) Explain Flynn's classification with the help of appropriate example.

   (b) Compare control flow computers to data flow computers.

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3. (a) What do you mean by virtual memory ? Explain.

   (b) Discuss the various schemes for main memory addressing.

4. Draw and explain architecture of star-100 or TI–ASC machine ?

5. (a) Compare between linear pipe-line and non-linear pipe-line.

   (b) Explain the following SIMD interconnection network : http://www.mjpruonline.com

   (i) Illiac

   (ii) Cube

6. (a) Explain the following :

   (i) UMA

   (ii) MUMA

   (iii) COMA

   (iv) Multiport memory.

   (b) What are the various methods to achieve concurrency for multiprocessing ?

7. Write short notes on (any *four*) :

   (i) RISC Scaler processor

   (ii) Window Register concept

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(iii) Cross bar switches

(iv) Modified Omega SIMD processor

(v) Multiprocessor operating system

(vi) Vector supercomputers.

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